

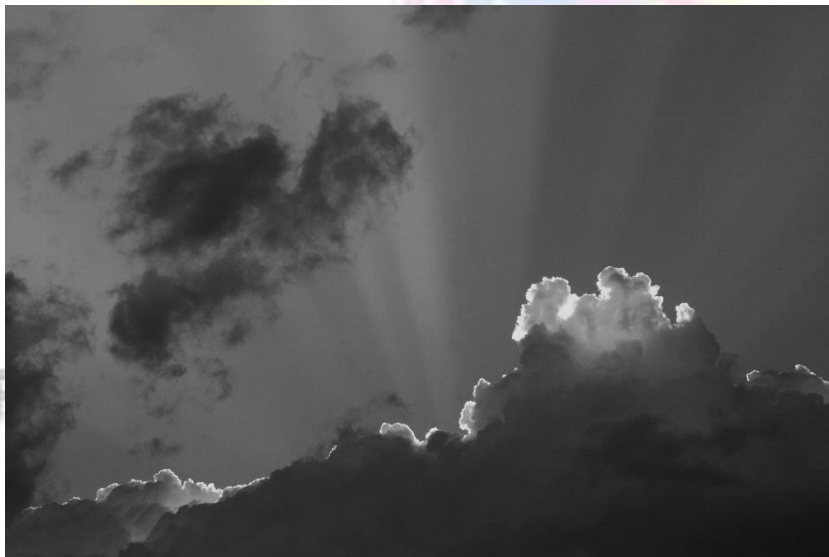
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# Science

Quarter 1– Module 3

C O L L O I D S



**Science– Grade 6**  
**Quarter 1 – Module 1: Colloids**

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**Describe the appearance and uses of homogenous and heterogenous mixtures**



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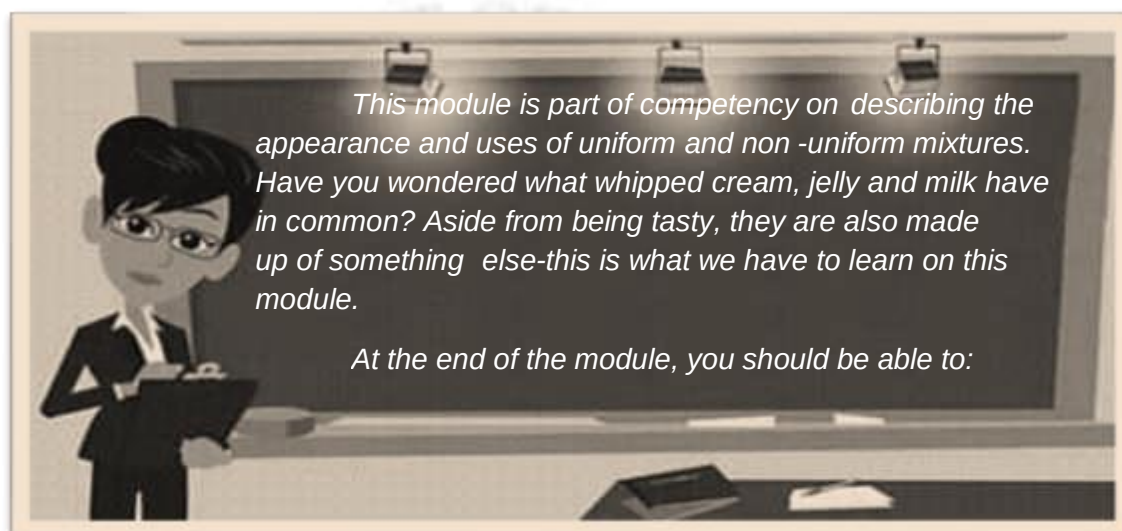
# Supplementary Learning Module for Elementary Learners

## LESSON

### COLLOIDS: HETEROGENEOUS MIXTURE



After accomplishing the tasks in your second module you learned about the characteristics of solution and suspension. That lesson also taught you how to identify and recognize common household materials of solution and suspension mixtures. This module will focus on colloid mixtures and its uses. You will also learn about the common colloidal materials found at home.



- infer colloids as a non-uniform mixture
- recognize common household colloid and their uses



**Directions:** Identify what is asked or described in each item then encircle the letter of the correct answer.

1. Which of the items below is a colloid?  
a. fruit salad   b. gelatin   c. lacquer
2. Marshmallow is costly, what type of food is marshmallow?  
a. suspension   b. solution   c. colloid
3. Which statement BEST describes Colloids mixture?  
a. colloids are homogeneous mixture  
b. the components of colloids are settled at the bottom  
c. particles of colloids cannot be seen by the naked eye
4. Natural colloid that is found inside the body \_\_\_\_\_.  
a. heart   b. blood   c. stomach

5. Ointment is an example of colloid, how is it used?
- a. condiment that enhances food flavor
  - b. for application on skin
  - c. for binding materials like cloth or paper

*Did you find the questions difficult? To check if your answer is correct look for the answer found in the last page of this module. Don't worry if you got a low score. This means you have more things to learn from this module.*



#### 4 PICS-ONE WORD GAME

**Directions:** The set of pictures describe the jumbled words below. Analyze the pictures carefully to find out the correct word.



|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| L | L | O | I | S | D | C | O |
|---|---|---|---|---|---|---|---|

Did you find the correct word? Yes, if your answer to the jumbled word is COLLOIDS, proceed to next activity to find out more about this lesson.



### INVESTIGATE MIXES!

**Directions:** Perform the activity below with the help of your parents or guardian. Record your answer in your activity notebook.

**Materials:**

drinking glass- 3pcs gaw-gaw  
or corn starch- 1 pack  
tap water and hot water  
penlight or flashlight  
sugar

- **Activity 1**
  1. Prepare one glass of tap water.
  2. Add a teaspoon of sugar. Stir the water until all sugar particles dissolve.
- **Activity 2**
  1. Stir a teaspoon of gaw-gaw in tap water.
  2. Pour the mixture into another glass of boiling water.
  3. Allow the mixture to cool and set aside.
- **Activity 3**
  1. Prepare another one glass of tap water.
  2. Add a teaspoon of uncooked gaw-gaw powder.
- **Activity 4**
  1. To each glass, direct a beam of light from a penlight in dark room. (Note: You may use a big black cloth to cover yourself in lieu of a dark room)
  2. Look at the ray of light focused at right angle with the mixture.
- **Questions:**
  1. Compare the colors of the mixtures.

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2. Which of the three mixtures is/ are homogeneous? Heterogeneous?

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3. Which is clear and transparent, or no particles are observed?

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4. Which has fine particles that do not settle? Are these particles visible to the naked eye?

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5. Which mixture has big particles that tend to settle at the bottom of the container?

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6. Which exhibits Tyndall Effect?

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• **Conclusion:**

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• **Guide questions.**

1. Which mixture exhibits rays that scattered when a beam of light is passed through it? What kind of Test is applied to the mixture?

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2. So therefore, how do we know that this mixture is a colloid? Why?

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*You have successfully done your investigation by doing the activity. This means you are now ready for another activity. To explore more about this lesson, move on to the next activity.*

### SHOPPING FOR COLLOIDS

**Directions:** Read the problem and answer the questions that follow. Record your answer in your activity notebook.

• **Problem:**

You and your mother bought the following items from the supermarket: *mayonnaise, butter, brown sugar, milk, oil, salt, detergent powder, shampoo, canned goods, dishwashing liquid, toothpaste, vinegar, soy sauce, olive oil, and ketchup.*

- **Select 6 items that are colloids and describe its uses. Write your answers on the table below:**

| Colloid Items | Uses of Colloids |
|---------------|------------------|
|               |                  |
|               |                  |
|               |                  |
|               |                  |

|  |  |
|--|--|
|  |  |
|  |  |

- **Question:**

7. How are these colloids important to our daily life?

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*Congratulations for accomplishing the activities! Keep going and believe that you can do it. Please check the Rubric of the activities in the last page. Proceed to the next page of this module to answer your doubt and queries about the activities you have just accomplished!*



A mixture with particles evenly scattered in a dispersed medium without settling down is called **COLLOID**.

To investigate the colloidal property of this mixture, try to dissolve flour in water. This will result to a colloid while sugar dissolved in water is a solution. The flour particles can absorb, reflect and scatter light; therefore, a beam of light passing through the set-up was visible. This scattering of light is called Tyndall Effect. A picture of this is shown in the next page.

Tyndall Effect is seen as a beam of light in a colloid because there is a scattering of light when light beams pass through it due to the dispersed particles which absorb, reflect and scatter the light.

Colloid is a special kind of mixture that contains tiny particles that cannot be seen by the naked eye but visible through an ultra microscope. The tiny particles do not settle down. Instead, they move rapidly and collide with or bump each other. This motion of the tiny particles is called Brownian motion. In adding a powdered milk to a glass of warm water, you can form a cloudy mixture, but the milk particles do not settle down. The milk that you drink every day is called colloid.



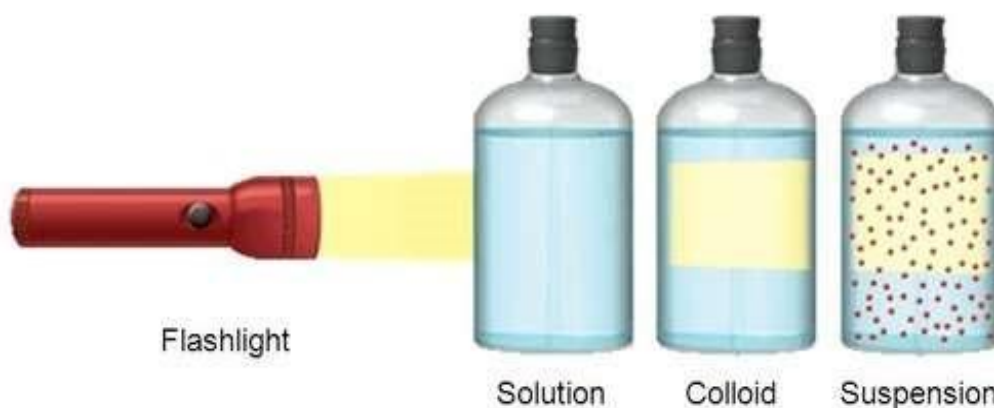
Solutions, suspensions and colloids exhibit characteristics that can be perceived as similar, but they are actually not. For instance, colloids have particles that are slightly smaller than the particles of a suspension; colloidal particles, however, are larger than the particles of solutions. Colloidal particles are in-between the sizes of suspension and solution particles and cannot be seen by the naked eye; instead, they are visible through a microscope only.

The different examples of colloids are important to daily life. Natural colloids such as blood, clouds and fog are basic for living things. Man-made colloids are also useful. Numerous colloids such as milk, butter, gelatin, jam, jelly, and other creamy substances such as mayonnaise and whipped cream, are used as food or ingredients for preparing food. Some colloids such as magnesium hydroxide, creams, and ointments are used as medicines and cosmetics. Paints have both protective and decorative functions. Styrofoam, inks, and white glues are used in offices and printing press. Insecticides are used in farming. Artificial aerosols are not environment friendly. These products contain chlorofluorocarbons (CFCs) which act as propellants. Experts say the CFCs, destroy the ozone layer, the layer that protects the earth from the ultraviolet rays of the sun.

### 15.3 Heterogeneous Aqueous Systems > Colloids

## The Tyndall Effect

The scattering of visible light by colloidal particles is called the **Tyndall effect**.



Source: Pearsoneducation.com

**For more information, watch this link from YouTube:**

<https://www.youtube.com/watch?v=Yo07oWNJgAA>

<https://www.youtube.com/watch?v=dvgWwqv5ZAY>



**Directions:** Read each statement. Encircle YES if it is correct, NO if it is incorrect then underline the word that makes the statement incorrect, then write the correct word.

- |        |                                                             |
|--------|-------------------------------------------------------------|
| YES NO | 1. Colloids are homogeneous mixtures.                       |
| YES NO | 2. Light cannot pass through colloidal particles.           |
| YES NO | 3. The components of a colloid do not settle at the bottom. |
| YES NO | 4. Colloid particles are bigger than suspension particles.  |
| YES NO | 5. Colloid particles are larger than solution particles.    |

*Congratulations, you have finished this module.*

*Kindly check your answers against the answer key found in last section of this module. Reflect and answer the assessment. If you are ready for the next module, you may proceed.*



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**Directions:** Write the use/s opposite each example of colloid. Record your answer in your activity notebook.

| Colloids           | Uses |
|--------------------|------|
| Shampoo            |      |
| Lotion             |      |
| Dishwashing liquid |      |
| Butter             |      |
| Cotton candy       |      |



**Directions:** Choose the letter of the best answer and write it in your notebook.

- \_\_\_1. Which of the following colloids is harmful to environment?  
a. whipped cream      b. artificial aerosol      c. jelly jam
- \_\_\_2. A phenomenon of scattering of light in colloidal mixture is called \_\_\_\_\_.  
a. Brownian motion      b. absorption      c. Tyndall effect
- \_\_\_3. Which of the items below is a colloid?  
a. cream      b. vegetable salad      c. lacquer
- \_\_\_4. Which colloid is used for medicines and cosmetics?  
a. magnesium hydroxide      b. paints      c. white glue
- \_\_\_5. Mixture of particles scattered evenly and dispersed without settling down.  
a. solute      b. colloid      c. solution
- \_\_\_6. Which statement BEST describes colloids mixture?  
a. the components of colloids are settled at the bottom  
b. particles of colloids can be seen by the naked eye  
c. colloids appear homogeneous mixture but are heterogenous mixtures.

\_\_\_7. It is a motion of tiny particles moving rapidly, colliding or bumping each other.

- a. Brownian motion    b. Tyndall effect    c. suspension

\_\_\_8. The following are colloidal properties of mixture EXCEPT:

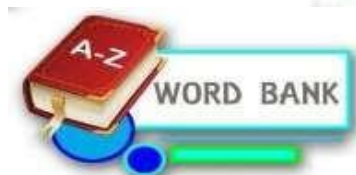
- a. it has Tyndall effect  
b. it has Brownian motion  
c. you can see with your naked eye

\_\_\_9. A natural colloid that is found inside the body \_\_\_\_\_.

- a. heart    b. stomach    c. blood

\_\_\_10. Ointment is an example of colloid, how is it used?

- a. condiment that enhances food flavor  
b. for application on skin  
c. for binding materials like cloth or paper



### Definition of Terms

**Colloids-** heterogeneous mixture with particles scattered in a dispersed medium without settling down.

**Brownian motion-** tiny particles moving rapidly, colliding or bumping each other.

**Tyndall effect-** light scattering phenomenon

**Chlorofluorocarbons (CFCs)-** consists of typical gases used in refrigerants and aerosol propellants, which are harmful to the ozone layer.

**Ozone layer-** layer of atmosphere that protects the earth from the ultraviolet rays of the sun

**Artificial aerosols-** are colloidal mixtures of sulfate, nitrate and carbon and particulate air pollutants.

## ANSWER KEY

### TRY THIS

1. b
2. c
3. a
4. b
5. b

### APPLY WHAT YOU HAVE LEARNED

1. NO
2. YES
3. YES
4. NO
5. YES

### ASSESS WHAT YOU HAVE LEARNED

1. b
2. c
3. b
4. a
5. b
6. c
7. a
8. c
9. c
10. b

### Scoring Rubric-Investigate Mixes!

| Point | Competency                                                                                             |
|-------|--------------------------------------------------------------------------------------------------------|
| 5     | Shows clear understanding of the topic and uses supports that are fully developed to probe the answer. |
| 4     | Shows understanding of the topic and uses supports that are developed to probe the answer.             |
| 3     | Shows a partial understanding of the topic and uses at least one support to probe the answer.          |
| 2     | Shows minimal understanding of the topic and there is no support that can probe the answer.            |
| 1     | Answer is completely irrelevant.                                                                       |
| 0     | No answer at all.                                                                                      |

## REFERENCES

Cover Art: [https://commons.wikimedia.org/wiki/File:Tyndall\\_Effect.jpg](https://commons.wikimedia.org/wiki/File:Tyndall_Effect.jpg)

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Google pictures

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